# Zaid Abu-Abbas

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## Education

**Bachelor of Science, Computer Science.** **Graduation Date: May 2026**
The George Washington University: Washington, D.C. **GPA: 3.60**

**Relevant Coursework:** Data Structures, Algorithms, Operating Systems, Databases, Machine Learning, Cloud Computing

## Technical Skills

**Languages:** Python, Java, C/C++, JavaScript, SQL, R, x86 Assembly, CSS, Node, Rust
**Technologies:** FastAPI, React, Node.js, AWS, Docker, Kafka, MongoDB, MySQL, Databricks, Grafana, ROS/ROS2, Linux, GitHub

## Professional Experience

**Cybersecurity and Anomaly Detection in Autonomous Drones** **October 2023 - May 2026**
Research in collaboration with Dr. Sibin Mohan, The George Washington University

- Configured PX4 flight controllers and integrated Vicon motion capture for precise tracking and validation
- Developed data pipelines to stream and preprocess flight and sensor data, and performed penetration testing/fault injection to evaluate hardware resilience
- Integrated a neural network anomaly detector within PX4's EKF real-time pipeline, optimizing inference to execute within the EKF update cycle without blocking flight state estimation

**Robotics Control Engineer Intern** **January 2026 - Present**
Temple Allen Industries, Rockville, MD

- Developed control logic for pneumatic robotic systems (EMMA™ and SAM) to automate sanding, grinding, and surface preparation tasks
- Integrated sensor-driven feedback (pressure sensors, DAQ systems) to improve precision, reliability, and safety in industrial workflows
- Tuned control parameters and pneumatic systems (regulators, tubing) while building and testing circuit prototypes with C/Python validation tools to ensure stability, accuracy, and operator ergonomics

**Software Engineering Intern - Machine Learning & Distributed Systems** **June 2025 - August 2025**
Data Surge, Remote

- Built a real-time hospital monitoring system with FastAPI, Python, and Kafka, reducing hazard detection from 20 minutes to under 5 seconds
- Developed an ML infection risk predictor (Random Forest, microbatching) for near real-time sensor data analysis and risk assessment
- Designed a fault-tolerant distributed system with 99%+ uptime and scalable configuration

## Projects

**NBA Game Outcome Predictor (ML + Real-Time Analytics Platform):** Web Application **October 2025 - May 2026**

- Engineered 40+ features (efficiency ratings, rolling metrics, Elo), achieving 76% pre-game accuracy on 2024–2025 data
- Developed SVM (pre-game) and XGBoost (live) models for dynamic, play-by-play predictions
- Designed a FastAPI backend and React frontend, containerized with Docker and deployed on AWS EC2

**ChatGPTravel – AI-Powered Optimized Travel Itineraries:** GitHub **September 2024 - May 2025**

- Built a web service that generates personalized travel itineraries based on user preferences (cost, ratings, travel time)
- Integrated generative AI and external APIs (Yelp, Foursquare) with optimization algorithms to produce efficient travel plans

## Publications

**Abu-Abbas, Z**., Ali, A., Faryadi, M., Surafel, Y., & Qu, X. (2026). "*Review-Informed Pilot Dataset and Baseline Modeling for NBA Game Winner Prediction.*" KDD 2026. **[Accepted]**

Kim, K. H., Narendran, T., Mehralizadeh, B., **Abu-Abbas, Z.,** Kara, D., Paruchuri, V., Mohan, S., Kimberly, G., Kim, J., & Eckhardt, J. *"Predictable by Design, Vulnerable by Nature: Security Consequences of Learnability in UAV State Estimators."* NDSS 2027. **[Under Submission]**